

TSA Meeting — Slide 3 Study Note

Research Gaps & Objectives

6 Identified Gaps · 6 Research Objectives · Prior Art Positioning
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Slide Overview and Narrative

What This Slide Shows

Slide 3 is the **research positioning** slide. It answers the two most important questions a TSA committee will ask: *What has not been done before?* (identified gaps) and *What exactly are you doing?* (research objectives). Every claim of novelty in the thesis is traceable to one of the six gaps on the left; every experimental chapter in the thesis delivers one of the six objectives on the right.

The **six gaps** can be grouped into three clusters:

- **Standards and modelling gap** (G1, G2): No relay standard handles real-time k_{ibr} ; grid-forming IBR protection-control coupling is unstudied.
- **Algorithmic gap** (G3, G4): No unified inverse-estimation framework exists; existing digital twins lack formal bounded non-blocking update guarantees.
- **System-level gap** (G5, G6): WAPC designs ignore IBR-induced coherency breakdown; adversarial ML attacks on protection are unmitigated.

The **six objectives** map one-to-one to these clusters: O1–O2 address the standards/modelling and algorithmic gaps; O3–O4 address the digital twin and ML gaps; O5–O6 address the system-level and security gaps. The slide is deliberately structured as a symmetrical 6-to-6 map.

Gap-to-Objective Master Map

Gap	Statement	Objective	How It Closes th
G1	No standard accounts for real-time k_{ibr}	O1: Unified SAMBP LM framework	LM identifies k_{ibr} on all relay types
G2	GFM protection-control coupling unstudied	O2: Self-adaptive modules (incl. microgrid)	Microgrid module L GFL/GFM dual-mo
G3	No unified inverse-estimation across relays	O1 (unified framework)	One LM solver element-specific mo
G4	DT lacks bounded non-blocking updates	O3: Non-blocking EKF digital twin	EKF + bounded law + non-blocking
G5	WAPC ignores IBR coherency breakdown	O5: WAPC with topology-aware IBR coherency	WAPC uses k_{ibr} in herency checks
G6	Adversarial ML attacks unmitigated	O4: Physics-informed ML; O6: Cybersecurity layer	PINN loss + GOOSE authentication

The Six Gaps: Why Each Is Genuinely a Gap

G1 — No Relay Standard Accounts for Real-Time IBR Admittance k_{ibr}

Standards Gap: IEC 60255 and IEEE C37 Assume Fixed Fault Current

The two governing relay standards are:

- **IEC 60255-151** (overcurrent relay characteristics): Defines ITOC/IDMT curves with fixed pickup I_p and time multiplier TMS. No provision for real-time pickup adaptation based on source composition.
- **IEEE C37.112** (inverse-time relay characteristics): Similarly assumes the fault current model is determined at commissioning.
- **IEC 61869** (instrument transformer standards): Specifies CT accuracy classes but does not address the distortion of fault current by IBR current limiting.

The standards were written when the only adjustable parameters were I_p , TMS, and zone reach — all set offline by a protection engineer. The concept of a relay that updates its own settings every 20 ms based on a real-time k_{ibr} estimate does not appear in any relay standard.

Industry gap: Even the latest “adaptive relay” products (Siemens SIPROTEC 5, ABB REL 670) use “setting groups” — a finite set of pre-configured offline settings that are switched based on topology events. They do not compute relay settings from first-principles model identification during the fault itself.

G2 — GFM Protection-Control Coupling Unstudied

Grid-forming (GFM) inverters operate as voltage-controlled sources emulating a virtual synchronous machine. Unlike GFL inverters, the GFM fault current is determined by its virtual impedance $Z_v = R_v + j\omega_0 L_v$:

$$I_{sc}^{GFM} = \frac{|V_0|}{|Z_v|} \in [1.5, 2.5] I_{rated} \quad (1)$$

The protection challenge: the same Z_v that limits fault current also participates in the microgrid’s voltage regulation and frequency droop. Changing Z_v to improve protection performance (e.g., raise fault current for relay visibility) directly compromises the GFM’s control stability. This *protection-control coupling* is documented in the power electronics literature (Taul 2020, Paquette 2015) but no protection relay design has explicitly addressed the constraint.

The SAMBP microgrid module (Ch 5) operates within the GFM control constraints by adapting *relay settings*, not inverter parameters — this is the only design approach that does not require changes to the GFM firmware.

G3 — No Unified Inverse-Estimation Framework Across Relay Types

Prior work addresses each relay type in isolation:

- Adamiak (2018): Adaptive 87L using online slope estimation for HVDC lines — specific to DC sources, not GFL/GFM.
- Hooshyar (2019): Retuned OC settings for high-IBR feeders — offline process, not inverse estimation during fault.

- Sadeghi (2021): Distance relay reach correction for IBR networks — element-specific correction table, no model identification.

Each paper proposes a *different* estimation method tailored to *one* relay type. There is no published work that: (a) defines a single mathematical problem class (bounded nonlinear least-squares) that covers 87L, OC, 87T, and distance simultaneously; and (b) proves convergence and safety properties for the unified solver. Chapter 2 fills this gap.

G4 — Digital Twin Approaches Lack Bounded Non-Blocking Update Laws

Existing power system digital twins (Daniluk 2021, IEEE PES DT task force 2022) use one of two approaches:

- **Offline batch estimation:** Run optimisation after the event, update settings during the next maintenance window. Latency: hours to days. Protection cannot benefit from the estimate in real time.
- **Online blocking estimation:** Run the estimator in the relay’s main compute loop. If the estimator takes >1 ms, the relay misses its trip decision window. No published protection DT has formal non-blocking guarantees.

Neither approach provides: (a) a formal proof that settings stay within safe bounds regardless of estimation quality, or (b) an asynchronous (non-blocking) design where the estimator runs in a background thread and the relay always has a valid (if possibly stale) parameter estimate. Chapter 9 provides both.

G5 — WAPC Designs Ignore IBR-Induced Coherency Breakdown

Classical WAPC coherency grouping (IEEE 1110, WECC stability criteria) clusters generators by *power angle synchronism*: machines whose rotor angles swing together in response to a disturbance form one coherent group. This grouping determines which relays are coordinated together.

IBRs have no rotor angle. Their “coherency” is defined by the PLL tracking bandwidth and the reactive current injection priority: two IBRs at opposite ends of a feeder may respond to the same voltage dip in diametrically opposite ways (one injects reactive, the other active), creating an *anti-coherent* pair.

Published WAPC designs (Apostolopoulos 2020, Sun 2022) apply SG coherency algorithms to IBR-dominated networks, producing incorrect relay groups. The SAMBP WAPC (Ch 7) uses the EKF-estimated k_{ibr} as a coherency proxy, grouping relays by their IBR penetration level rather than by power angle synchronism.

G6 — Adversarial ML Attacks on Protection Relays Unmitigated

The adversarial ML literature (Goodfellow 2015, Madry 2018) demonstrates that small crafted perturbations can cause neural network classifiers to misclassify with high confidence. In protection, this means:

- $A \leq 5\%$ perturbation in a CT secondary current can cause an unconstrained NN to classify a genuine fault as normal operation (miss trip).
- $A \leq 3\%$ perturbation can cause classification of load current as a fault (false trip).

The protection literature has not published a mitigation for this attack class. The SAMBP response (Ch 6 and Ch 8) is two-pronged: (a) the PINN loss function embeds KVL/KCL constraints, so any perturbation that causes misclassification must also violate the power flow equations — raising the minimum attack energy; and (b) HMAC-SHA256 on GOOSE prevents the attacker from injecting false measurements via the communication network.

The Six Objectives: Measurable Success Criteria

How Each Objective Is Verified

The TSA will ask: *how do you know you have achieved each objective?*

Obj.	Statement	Success Criterion (quantified)
O1	Unified SAMBP LM framework	Convergence $R^2 > 0.999$ in <20 iterations for all models; Theorem 2.1 convergence proof
O2	Self-adaptive protection modules for 87L, SyncOC, 87T, 21, microgrid	TPR = 1.000; FPR = 0.000; avg. trip time ≤ 80 ms across all elements
O3	Non-blocking EKF digital twin	$\hat{k}_{ibr}$ RMSE ≤ 0.024 pu; EKF background thread latency ≤ 2 ms; relay decision not blocked
O4	Physics-informed ML for fault classification	Classification accuracy $>95\%$ under $\varepsilon = 5\%$ CT perturbation; KVL/KCL residual $< 10^{-4}$ pu
O5	Wide-area coordinator with IBR coherency	Fault localisation accuracy 100%; N-1 verified; settings dispatch <100 ms
O6	Cybersecurity layer	Spoofing success $< 0.001\%$; GOOSE latency overhead < 1 ms; risk reduction $>50\%$ per IEC 62351 matrix

Prior Art Analysis: Three Competing Approaches and Their Limits

How SAMBP Differs from Existing Approaches

Approach A — Legacy Relay Retuning (Hooshyar 2019, Sadeghi 2021):

Premise: Compute new fixed settings that are “better” for an IBR-heavy network; apply them during commissioning or scheduled maintenance.

Limitation: The setting is still fixed. As IBR fleet composition changes between maintenance cycles (seasonal wind variation, solar outages, new capacity additions), the setting becomes stale. At $k_{ibr} = 0.1$ vs. $k_{ibr} = 0.9$, the optimal bias slope differs

by $4\times$; no single fixed setting covers both.

SAMBP difference: Real-time LM estimation tracks k_{ibr} continuously; settings update every fault cycle.

Approach B — Communication-Assisted Schemes (Gers 2011, Zhang 2020): *Premise:* Use optical-fibre 87L (full differential) or POTT to eliminate the need for fault current magnitude thresholds.

Limitation: These schemes *still have* a bias slope k_m . At $k_{ibr} < 0.2$, the fixed $k_m = 0.30$ causes a missed trip even with full communication (as demonstrated in Ch 3 TR-08 results). Communication latency under cyberattack or fibre cut disables the scheme entirely.

SAMBP difference: Adaptive $k_m^*(k_{ibr})$ works at all k_{ibr} values; Stage-2 gate handles latency gracefully.

Approach C — Pure Machine Learning (Jamali 2020, Liao 2022):

Premise: Train a deep neural network on simulation data covering all fault types and IBR penetration levels; deploy as the relay decision function.

Limitation: (a) Produces physics-impossible outputs (e.g., classifying a fault where KVL is violated by the prediction); (b) generalisation to out-of-distribution IBR types (e.g., DFIG crowbar activation) is poor; (c) adversarial perturbation vulnerability is unaddressed in every published protection ML paper.

SAMBP difference: PINN loss embeds KVL/KCL; the LM model is physics-first; adversarial hardening via HMAC + physics constraints.

Thesis Table 1.1: Summary of Original Contributions (Authoritative)

Table 1.1 from the Thesis (Chapter and Result Numbers Are Final)

ID	Ch	Contribution	Status
C1	3	Physics-constrained adaptive line differential (87L) protection with IBR-aware threshold $k_m^*(k_{ibr})$, achieving TPR = 1.000 and zero false trips across the full IBR penetration range	[P]
C2	4	Levenberg–Marquardt inverse-estimation framework for synchronous generator over-current protection, with bounded adaptive update law and four-component confidence gating ($R^2 > 0.999$)	[P]
C3	6	Non-blocking EKF digital twin for real-time joint estimation of IBR penetration ratio k_{ibr} , line impedance, and IBR admittance from IED phasor measurements (RMSE ≤ 0.024 pu)	[P]
C4	7	Wide-area protection coordinator with PMU-based fault localisation, N-1 security verification, and automatic CTI-graded relay settings dispatch via IEC 61850 GOOSE	[P]

[P] = validated with full numerical simulation results.

Correct chapter mapping for all cross-references: Ch 2 = Mathematical

Foundations (LM framework) · Ch 3 = 87L Line Differential (C1) · Ch 4 = SyncOC Generator OC (C2) · Ch 5 = Adaptive Distance Protection · Ch 6 = EKF Digital Twin (C3) · Ch 7 = WAPC (C4) · Ch 8 = Cybersecurity

Novelty Claims: What SAMBP Does That No Prior Work Has Done

Three Novelty Claims

The thesis positions SAMBP as the **first work** to do all three of:

1. **Treat adaptive protection as a unified inverse estimation problem** with explicit model identification across all element types (87L, OC, 87T, distance). Prior work treats each element in isolation with different estimation methods. The SAMBP LM framework provides one solver, one confidence gate ($\gamma \geq \gamma_{th}$), and one convergence proof (Theorem 2.1) for all elements.
2. **Integrate local relay adaptation, digital-twin state estimation, and centralised coordination** into a single closed-loop self-adaptive system. Prior works on adaptive protection, digital twins, and WAPC are published independently with no shared interface specification. SAMBP defines the three-tier architecture (Tier 1 EKF oracle $\rightarrow$ Tier 2 protection modules $\rightarrow$ Tier 3 WAPC) with formal data contracts between tiers.
3. **Provide formal confidence-gating and bounded update laws** that guarantee relay *safety* under estimation uncertainty. Existing adaptive relay papers show performance improvement without proving that a wrong estimate cannot push settings to an unsafe value. The bounded update (Proposition 2.1) and Stage-2 gate provide this guarantee.

Gap and Objective Mapping to Thesis Structure

Gap	Obj.	Ch	Contribution	Key Result
G1, G3	O1	2	Unified LM inverse-estimation framework	Theorem 2.1; $R^2 > 0.999$; ≤ 20 iter
G1, G2	O2	3–5	Self-adaptive protection modules (87L C3, SyncOC C4, Dist C5)	TPR = 1.000; FPR = 0.000
G4	O3	6	Non-blocking EKF digital twin (C3)	RMSE ≤ 0.024 pu; 2 ms thread la- tency
G6	O4	6	Physics-informed NN (PINN) classifier	$> 95\%$ ac- cu- ra- cy at $\varepsilon = 5\%$
G5	O5	7	WAPC with IBR-aware coherency (C4)	100% lo- cal- isa- tion; 54.9% risk re- duc- tion
G6	O6	8	Cybersecurity: HMAC + adversarial PINN	Spoofing $< 0.001\%$; over- head 0.72 ms

Cross-Thesis Connections

Every subsequent slide in the TSA presentation fills in one row of the table above. The logical flow is:

- **Slide 5** (Ch 2 LM framework) $\leftarrow$ closes G3 and G4.
- **Slide 6** (Ch 3 IBR characterisation) $\leftarrow$ quantifies G1 and G2 with the failure map table.

- **Slides 7, 8** (Ch 3, C1: 87L) ← delivers O2 for 87L.
- **Slide 9** (Ch 4, C2: SyncOC) ← delivers O2 for OC.
- **Slide 11** (Ch 5: Distance) ← delivers O2 for distance.
- **Slide 4** (Three-Tier Architecture) ← shows O1–O6 integrated.

Slide 3 (this study note) is the **gateway** slide: the TSA should have it clear in mind when evaluating every technical slide that follows. If a committee member challenges the novelty of any result, the answer traces back to one of the six gaps stated here.

Key Reference Points for Literature Defence

Standards and Papers to Name-Drop

Reference	What It Says and How SAMBP Differs
IEC 60255-151	Defines fixed-curve OC protection; no provision for real-time pickup adaptation
IEEE C37.112	Standardises IDMT relay characteristics; no IBR model
Hooshyar (2019)	IBR feeder OC retuning; offline process; no LM estimator
Sadeghi (2021)	Distance relay reach correction for IBR; element-specific table; no unified framework
Adamiak (2018)	Adaptive 87L for HVDC; DC-source specific; not GFL/GFM
Goodfellow (2015)	Adversarial examples in ML; SAMBP applies this to protection relays and mitigates it
Jamali (2020)	DNN fault classifier for IBR networks; no physics constraints; adversarial not addressed
IEEE PES DT TF (2022)	Digital twin survey; no bounded update law; no non-blocking guarantee
Taul (2020)	GFM protection-control coupling analysis; no relay algorithm proposed

Timed Speaker Script (3 Minutes)

What to Say at the TSA Meeting

[0:00–0:30] Setting the Scene

“Slide 3 identifies the six specific gaps that this thesis fills. I grouped them into three clusters. The first two are standards and modelling gaps: no relay standard in IEC 60255 or IEEE C37 accounts for real-time IBR admittance, and the protection-control coupling unique to grid-forming IBRs has not been addressed in any relay design. These two gaps mean that even if a protection engineer wanted to build an adaptive relay today, there are no published methods to follow.”

[0:30–1:10] Algorithmic and DT Gaps

“The next two are algorithmic gaps. There is no unified inverse-estimation framework: the OC literature has one method, the 87L literature has a different method,

and distance has a third. None of them share a common solver, confidence gate, or convergence proof. The SAMBP Chapter 2 provides that unified framework. The digital twin gap is more specific: every published protection digital twin either runs offline or runs blocking in the relay's main thread. Neither has a formal proof that settings stay within safe bounds during a bad estimate. My bounded update law and Stage-2 gate fill this gap."

[1:10–1:50] System and Security Gaps

"The last two are system-level gaps. Wide-area protection coordinators today group relays by power-angle coherency — a concept that only applies to synchronous generators. IBRs have no rotor angle. My WAPC groups relays by IBR penetration ratio instead, using the EKF estimate as the coherency proxy. And the adversarial ML gap: it has been shown in computer vision that a 5% perturbation in the input can fool a neural network. For a protection relay, that means a 5% injected CT error causes a misclassification. No protection ML paper has addressed this. My PINN loss function and HMAC countermeasure address it."

[1:50–2:30] The Six Objectives

"The six research objectives on the right close the six gaps one-to-one. The unified SAMBP framework closes G1 and G3. Self-adaptive modules close G1 and G2 across all five protection element types. The non-blocking EKF digital twin closes G4. Physics-informed ML closes G6's classifier aspect. The wide-area coordinator closes G5. And the cybersecurity layer closes G6's communication aspect. Each objective has a quantified success criterion: TPR = 1.000, FPR = 0.000, RMSE ≤ 0.024 pu, spoofing below 0.001%."

[2:30–3:00] Novelty Position

"I want to be explicit about what is new. This is the first work to treat adaptive protection as a single unified inverse estimation problem across all relay types. The first to integrate local adaptation, digital twin, and centralised coordination into one closed-loop system. And the first to provide formal bounded-update and confidence-gate guarantees that prove safety under estimation uncertainty. Prior adaptive relay work improves performance; this thesis guarantees it."

Anticipated Q&A — Five Likely Committee Questions

Q1: How do you know no relay standard accounts for real-time k_{ibr} ? What about IEC 61968 or IEC 62351?

Answer: IEC 61968 is a CIM (Common Information Model) standard for utility data exchange — it defines data schemas, not relay algorithms. IEC 62351 is the cybersecurity series for IEC 61850 communications — it also does not define relay algorithms. The relay algorithm standards are IEC 60255-151 (overcurrent), IEC 60255-121 (distance), and IEEE C37.112 (inverse time); none of these define a mechanism for real-time relay threshold computation from an online k_{ibr} estimate. I have reviewed all three standards in Chapter 1's literature review. The closest is IEC 60255-121 Annex B, which allows for "special characteristics" by mutual agreement between manufacturer and user — this is essentially a blank cheque that

acknowledges the standard is insufficient, which is exactly the gap.

Q2: What specifically is “IBR-induced coherency breakdown” in G5?

Answer: Classical coherency in power systems means that during a disturbance, a group of synchronous generators swing together in power angle — their rotor speeds stay nearly identical, so they form one “coherent group” for protection coordination. When you replace SGs with IBRs, coherency breaks down in two ways: (a) IBRs have no rotor — they have no inertia and no swing equation; (b) IBRs actively inject current in a direction commanded by the LVRT control law, not by physics. Two IBRs at opposite ends of a feeder can inject reactive current in *opposite* directions simultaneously, creating an anti-coherent pair. The WAPC must know which relays are protecting IBR-dominated lines to apply different coordination criteria. The SAMBP WAPC uses $\hat{k}_{ibr}$ from the EKF as a coherency proxy: relays with similar $\hat{k}_{ibr}$ values are grouped together for CTI coordination.

Q3: “Protection-control coupling of GFM IBRs unstudied” seems strong. Are there no papers on this?

Answer: The analysis of GFM protection challenges exists in the power electronics literature (Taul 2020, Lasseter 2020). What is *unstudied* is a relay algorithm that explicitly accounts for the coupling. Taul 2020 shows that changing the virtual impedance Z_v to increase fault current for relay visibility degrades the GFM’s voltage regulation stability — but proposes no relay solution. The SAMBP microgrid module (Ch 8) is the first relay algorithm that leaves Z_v untouched and adapts the relay settings instead, operating entirely within the GFM’s existing control constraints. So the gap is precisely stated: the algorithm has not been published, even though the problem has been identified.

Q4: You have 6 gaps and 6 objectives but the thesis has 4 principal contributions (C1–C4). How do these reconcile?

Answer: The six objectives on the slide are *granular*; the four principal contributions (C1–C4) are *consolidated* for thesis chapter organisation: From Table 1.1 of the thesis: C1 (Ch 3) is the adaptive 87L line differential achieving TPR = 1.000; C2 (Ch 4) is the LM inverse-estimation framework for SyncOC with $R^2 > 0.999$; C3 (Ch 6) is the non-blocking EKF digital twin with RMSE ≤ 0.024 pu; C4 (Ch 7) is the WAPC with PMU fault localisation and CTI-graded settings dispatch. All four are validated [P]. The six-objective structure on the slide unpacks the four chapters into finer deliverables (e.g., O4 = PINN and O6 = cybersecurity are both inside Ch 6/Ch 8 respectively), making the novelty mapping explicit for the committee.

Q5: Are there commercial products (ABB, Siemens, GE) that already do adaptive protection? How is your work different?

Answer: Yes — ABB REL 670, Siemens SIPROTEC 7SA87, and GE Multilin D90 all claim “adaptive” protection. Their adaptation mechanism is *setting groups*: the relay engineer pre-configures 2–8 groups of fixed settings, each for a different

expected network condition, and an external trigger (SCADA topology message or digital input) switches between groups. This is *not* real-time inverse estimation: the groups are computed offline, and the relay does not identify any parameter from its own CT measurements. It is essentially “manual look-up table” adaptation. SAMBP is fundamentally different: the relay computes its own setting from a real-time physical model fit. The analogy: commercial “adaptive” relays are like a photographer choosing between pre-set exposure modes (portrait, sport, night); SAMBP is like a camera that measures the actual scene lighting and computes the correct exposure.

Summary for the TSA Committee

Six identified gaps (two standards, two algorithmic, two system-level) map to six research objectives, each with a quantified success criterion. SAMBP’s novelty rests on three claims: unified inverse-estimation across all relay types, closed-loop integration of adaptation and coordination, and formal safety guarantees via bounded updates and confidence gating. Every technical slide that follows delivers one of these six objectives.